Sequences — problem 9.10

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Let u_0, v_0, $u_0 < v_0$ be given. The sequences u_n, v_n are defined by $u_n = (u_{n-1} + v_{n-1})/2$, $v_n = (u_{n-1} + 2v_{n-1})/3$. Prove that both have the same limit L, $u_0 < L < v_0$.